# Lecture 14 Model Assessment: Cross-Validation

## STAT 441: Statistical Methods for Learning and Data Mining

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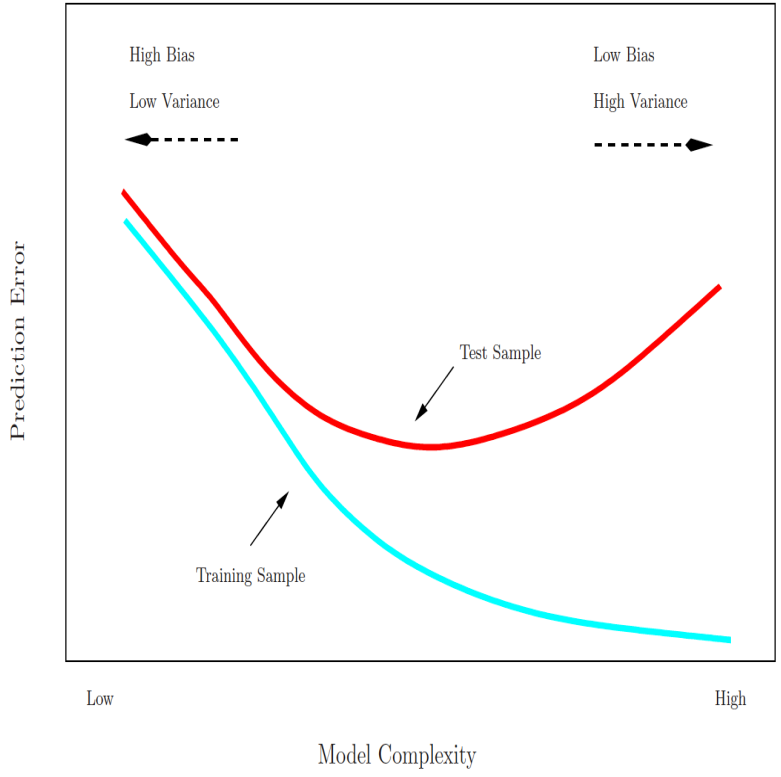
# Outline

# Training Error Versus Test Error

- The test error is the average error that results from using a statistical learning method to predict the response on a new observation, one that was not used in training the method.
- In contrast, the training error can be easily calculated by applying the statistical learning method to the observations used in its training.
- But the training error rate often is quite different from the test error rate, and in particular the former can dramatically underestimate the latter.

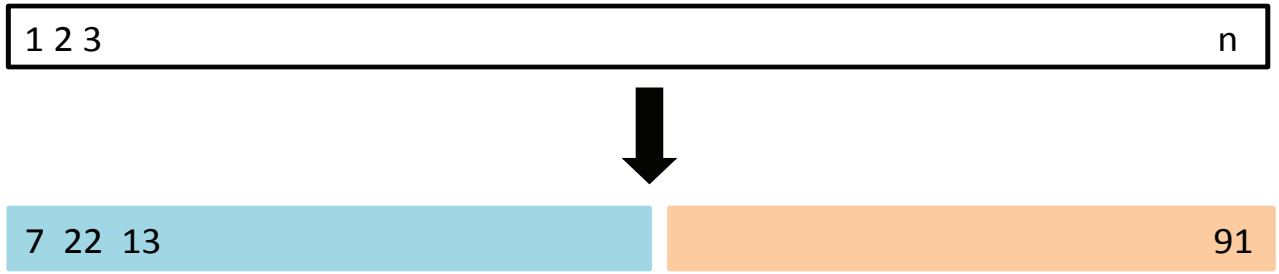


# Prediction Error Estimates

- Best solution: a large designated test set. Often not available.
- Some methods make a mathematical adjustment to the training error rate in order to estimate the test error rate. These include the $C_p$ statistic, AIC and BIC.
- Here we instead consider a class of methods that estimate the test error by holding out a subset of the training observations from the fitting process, and then applying the statistical learning method to those held out observations
- In particular, we randomly divide the available set of samples into two parts: a training set and a validation or hold-out set.

# Validation Set Approach

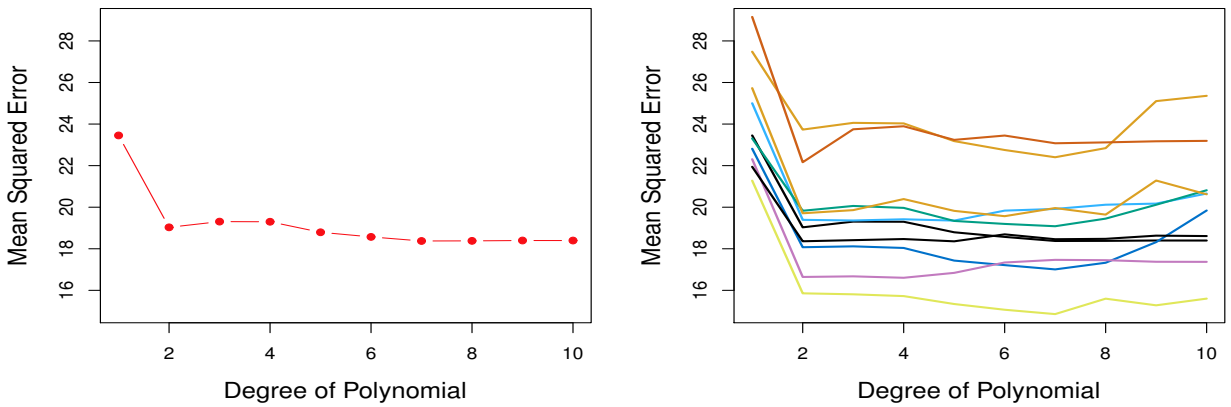
- The model is fit on the training set, and the fitted model is used to predict the responses for the observations in the validation set.
- The resulting validation-set error provides an estimate of the test error.
- This is typically assessed using MSE in the case of a quantitative response and misclassification rate in the case of a qualitative (discrete) response.
- A random splitting into two halves: left part is training set, right part is validation set.




- Split the data in two parts; Train the method in the first part; Compute the error on the second part.

# Automobile Data

- Want to compare linear vs higher-order polynomial terms in a linear regression.
- We randomly split the 392 observations into two sets, a training set containing 196 of the data points, and a validation set containing the remaining 196 observations.
- Left panel shows single split; right panel shows multiple splits.




- Estimates can vary considerably with different train-validation splits.
- Only subset of points used to evaluate model.

# Drawbacks of Validation Set Approach

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- The validation estimate of the test error can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the validation set.
- In the validation approach, only a subset of the observations — those that are included in the training set rather than in the validation set — are used to fit the model.
- This suggests that the validation set error may tend to overestimate the test error for the model fit on the entire data set.

# K-fold cross-validation

- K-fold cross-validation is a widely used approach for estimating test error.
- Estimates can be used to select best model, and to give an idea of the test error of the final chosen model.
- Idea is to randomly divide the data into $K$ equal-sized parts. We leave out part $k$, fit the model to the other $K-1$ parts (combined), and then obtain predictions for the left-out $k$-th part.
- This is done in turn for each part $k = 1, 2, \cdots, K$, and then the results are combined.
- Divide data into $K$ roughly equal-sized parts ($K = 5$ here)

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| Validation | Train | Train | Train | Train |

# K-fold cross-validation

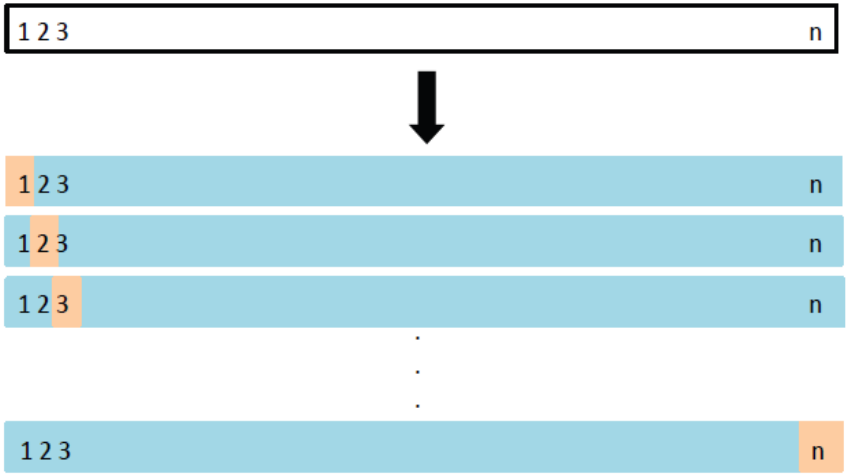
- Let the $K$ parts be $C_1, C_2, \cdots, C_K$, where $C_k$ denotes the indices of the observations in part $k$. There are $n_k$ observations in part $k$: if $N$ is a multiple of $K$, then $n_k = n/K$.
- Compute

$$\mathrm{CV}_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k,$$

  where $\mathrm{MSE}_k = \sum_{i \in C_k} (y_i - \hat{y}_i)^2 / n_k$, and $\hat{y}_i$ is the fit for observation $i$, obtained from the data with part $k$ removed.
- Setting $K = n$ yields $n$-fold or leave-one out cross-validation (LOOCV).

# Leave one out cross-validation

- For every $i = 1, \ldots, n$
  - train the model on every point except $i$
  - compute the test error on the held out point
- Average the test errors



# A nice special case!

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- With least-squares linear or polynomial regression, an amazing shortcut makes the cost of LOOCV the same as that of a single model fit!
- The following formula holds:

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

  where $\hat{y}_i$ is the $i$-th fitted value from the original least squares fit, and $h_{ii}$ is the leverage (diagonal elements of the hat matrix). This is like the ordinary MSE, except the $i$-th residual is divided by $1 - h_{ii}$.
- LOOCV sometimes useful, but typically doesn't shake up the data enough. The estimates from each fold are highly correlated and hence their average can have high variance.
- A better choice is $K = 5$ or 10.

# LOOCV v.s. K-fold cross-validation

**1. K-fold Cross-Validation**

Instead of one big split, you divide the data into $K$ equal groups (folds). You train the model $K$ times, each time leaving out a different group to use as the validation set.

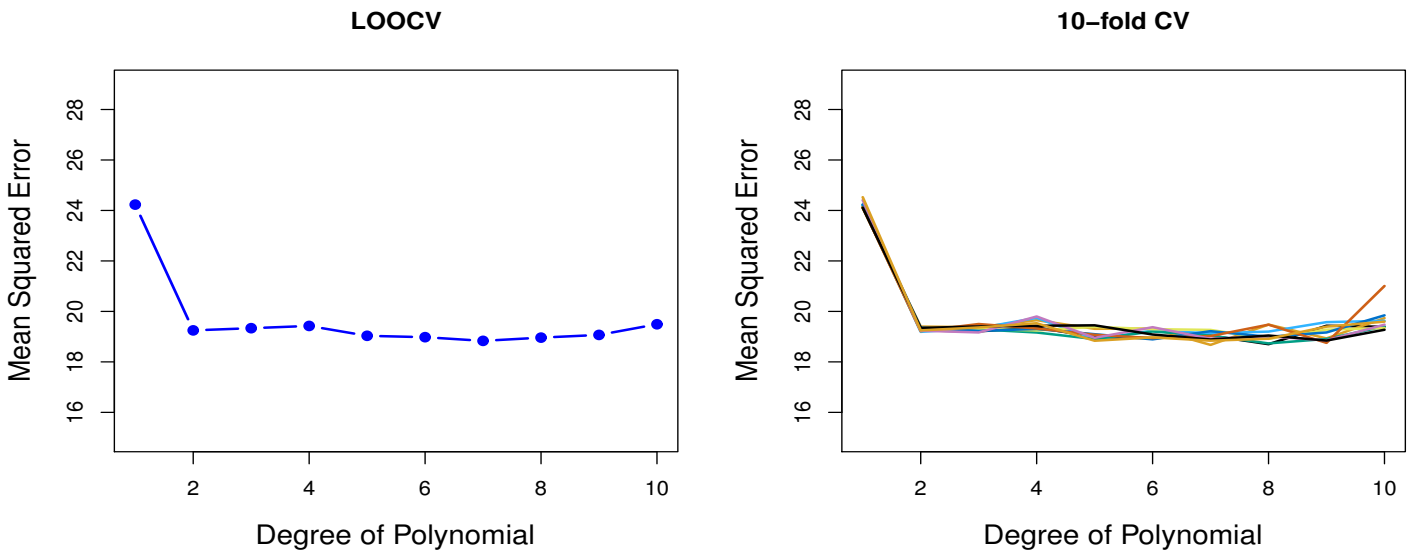
- **The Problem:** Because you are training on less data (e.g., in 5-fold, you only use 80% of the data each time), it introduces **bias**. The model doesn't see the full picture, so it tends to overestimate the true test error.
- **The Benefit:** It doesn't suffer from the wild swings of the "multiple random splits" you saw earlier.

**2. LOOCV (Leave-One-Out Cross-Validation)**

This is the extreme version where $K = n$. You leave out exactly **one** data point, train on everything else, and repeat this for every single point in the dataset.

- **The Benefit:** There is virtually no bias because you are using almost the entire dataset for training every time.
- **The Problem:** Because the training sets for each fold are almost identical (they only differ by one point), the outputs are highly correlated. This actually increases the **variance** of your final test error estimate.

- K-fold CV depends on the chosen split.
- In K-fold CV, we train the model on less data than what is available. This introduces bias into the estimates of test error.
- In LOOCV, the training samples highly resemble each other. This increases the variance of the test error estimate.
- $K = 5$ or 10 provides a good compromise for this bias-variance tradeoff.



# K-fold cross-validation for Classification Problems

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- Let the $K$ parts be $C_1, C_2, \cdots, C_K$, where $C_k$ denotes the indices of the observations in part $k$. There are $n_k$ observations in part $k$: if $N$ is a multiple of $K$, then $n_k = n/K$.
- Compute

$$\mathrm{CV}_K = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{Err}_k,$$

  where $\mathrm{Err}_k = \sum_{i \in C_k} I(y_i \neq \hat{y}_i)/n_k$, and $\hat{y}_i$ is the fit for observation $i$, obtained from the data with part $k$ removed.
- The estimated standard deviation of $\mathrm{CV}_K$ is

$$\widehat{\mathrm{SE}}(CV_K) = \sqrt{\sum_{k=1}^{K} \left(\mathrm{Err}_k - \bar{\mathrm{Err}}\right)^2 / (K-1)}.$$

- This is a useful estimate, but strictly speaking, not quite valid.

# An Example

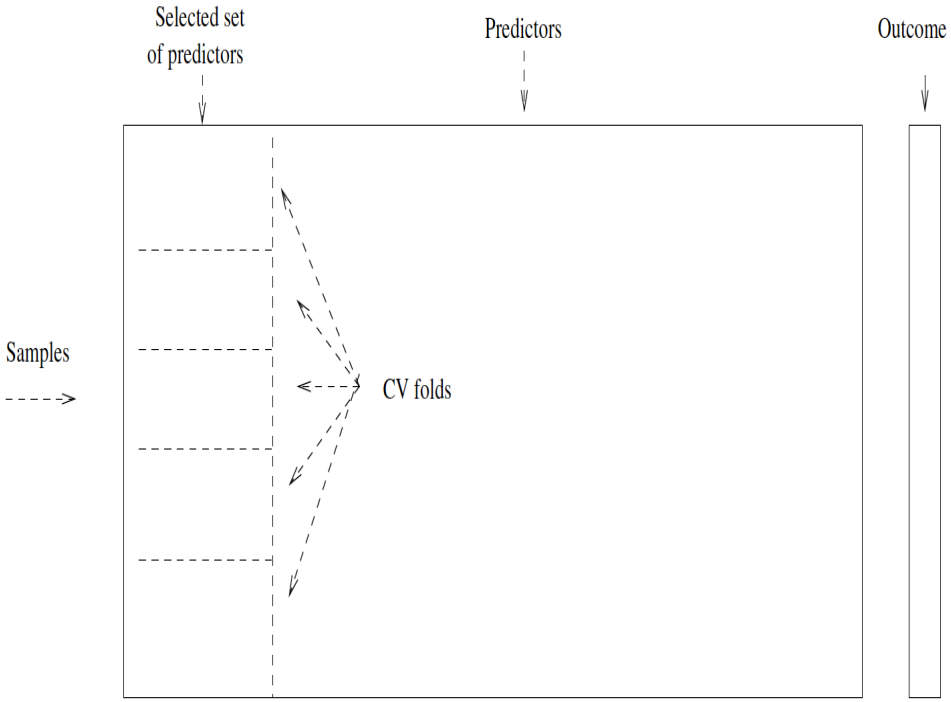
- Consider a simple classifier applied to some two-class data:
  - Step 1 Starting with 5000 predictors and 50 samples, find the 100 predictors having the largest correlation with the class labels.
  - Step 2 We then apply a classifier such as logistic regression, using only these 100 predictors.
- How do we estimate the test set performance of this classifier?
- Can we apply cross-validation in step 2, forgetting about step 1?

# No!

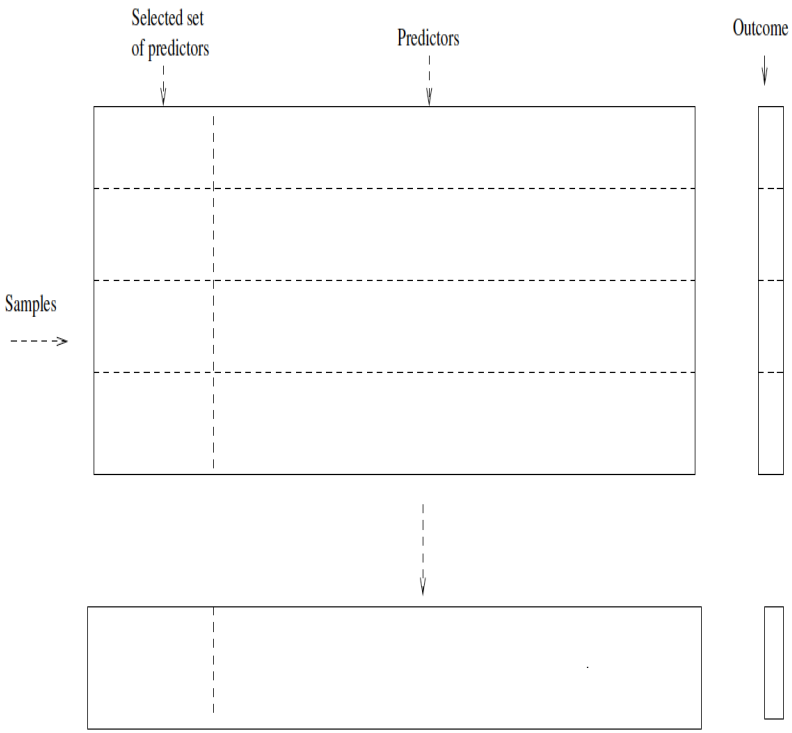
- This would ignore the fact that in Step 1, the procedure has already seen the labels of the training data, and made use of them. This is a form of training and must be included in the validation process.
- It is easy to simulate realistic data with the class labels independent of the outcome, so that true test error =50%, but the CV error estimate that ignores Step 1 is zero! Try to do this yourself.
- We have seen this error made in many high profile genomics papers.
- Wrong: Apply cross-validation in step 2. Right: Apply cross-validation to steps 1 and 2.

# Wrong Way and Right Way

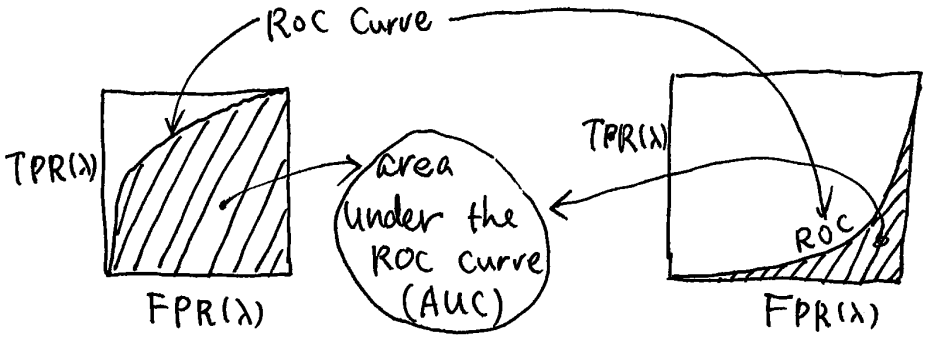
## Wrong Way




## Right Way

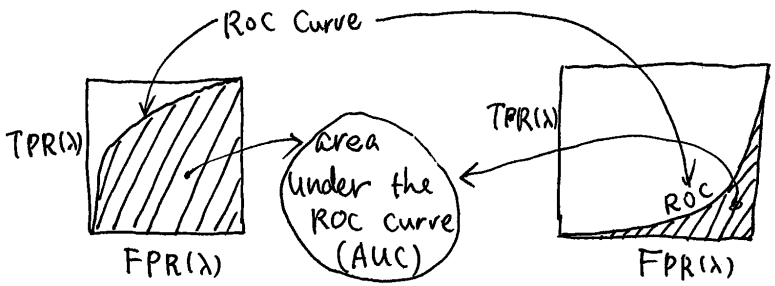


# The Area Under ROC Curve (AUC)




- AUC is defined as the area under the ROC curve. So an AUC score ranges from 0 to 1.
- AUC could be used to evaluate the performance of a classifier.
- AUC socre close to 1 indicates a good classifier.
- AUC close to 0 indicates a good classifier yet one should use it reversely.
- AUC close to 0.5 indicates a bad classifier. In particular, AUC$\approx$ 0.5 means the classifier has nearly the same performance like a random guess. Note that a random guess is also a classifier, although with poor performance.

# Computing AUC




$$AUC = \int_0^1 \frac{TP}{AP} d\frac{FP}{AN} = \int_{-\infty}^{\infty} P(f(y) > \lambda | y \in AP) P(f(x) = \lambda | x \in AN) d\lambda$$

$$= \frac{\int_{-\infty}^{\infty} P(f(y) > \lambda \text{ and } y \in AP) P(f(x) = \lambda \text{ and } x \in AN) d\lambda}{P(y \in AP) P(x \in AN)}$$

$$= \frac{P(y \in AP, x \in AN, \text{ and } f(y) > f(x))}{P(y \in AP) P(x \in AN)}$$

$$= \frac{\frac{1}{m^2} \sum_{x \in AN} \sum_{y \in AP} \mathbf{1}_{f(y) > f(x)}}{\frac{\#AP}{m} \times \frac{\#AN}{m}} = \frac{\sum_{x \in AN} \sum_{y \in AP} \mathbf{1}_{f(y) > f(x)}}{\#AP \times \#AN},$$

where $\mathbf{1}_{f(y) > f(x)} = 1$ if $f(y) > f(x)$ and $= 0$ otherwise.

With true labels available, either from oracle, or from cross-validation, AUC of a score-based classifier can be computed.

## Example

For the following classification output, compute the AUC.

| True Lable | Predicted Score |
|---|---|
| P | 1.7 |
| P | 8.1 |
| P | 7.8 |
| P | 9.3 |
| N | 6.2 |
| N | 6.1 |
| N | 1.2 |
| N | 2.9 |
| N | 5.8 |
| N | 6.3 |

# Computing AUC

With true labels available, either from oracle, or from cross-validation, AUC of a score-based classifier can be computed.

**Example**

For the following classification output, compute the AUC.

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| N | 2.9 |
| N | 5.8 |
| N | 6.3 |

f(y)

f(x)

| 9.3 | 7.8 | 8.1 | 1.7 | |
|---|---|---|---|---|
| ✓ | ✓ | ✓ | | |
| ✓ | ✓ | ✓ | | |
| ✓ | ✓ | ✓ | ✓ | |
| ✓ | ✓ | ✓ | | |
| ✓ | ✓ | ✓ | | |
| ✓ | ✓ | ✓ | | |

$\sum_x \sum_y 1_{f(y) > f(x)} = 19$

$\# AP = 4 \quad \# AN = 6$

$\therefore AUC = \frac{\sum_x \sum_y 1_{f(y) > f(x)}}{\# AP \ \# AP}$

$= \frac{19}{4 \times 6} = 0.7917$

# Summary and Remark

- Validation Set Approach
- K-fold Cross-Validation
- An Example
- Read textbook Chapter 5 and code